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# CausalLens: Diagnostics-First Causal Inference Software in Python

**Matthew Powell**
Independent researcher

**Abstract**

CausalLens is a Python package for causal effect estimation, diagnostics, sensitivity analysis, and manuscript-oriented reporting across observational and quasi-experimental designs. The package combines regression adjustment, propensity-score matching, inverse probability weighting (IPW), doubly robust (DR) estimation, cross-fitted doubly robust estimation, difference-in-differences (DiD), synthetic control, instrumental variables (IV), regression discontinuity (RD), manipulation testing, and bunching analysis in a shared diagnostics-first workflow. The contribution is not a new estimator or identification strategy. Instead, CausalLens provides a compact software layer that keeps overlap checks, balance summaries, sensitivity diagnostics, reference-parity checks, and exportable outputs close to the effect estimate. This positions the project as review-oriented statistical software: it helps users fit established causal estimators while retaining the assumption and validation context needed for replication and peer review. In the current replication bundle, Lalonde benchmark estimates fall between $1429 and $1710 across four estimators, while NHEFS smoking-cessation estimates fall between 3.26 and 3.33 kg, illustrating that the package reproduces recognizable benchmark ranges while exporting the associated diagnostics.

*Keywords*: causal inference, statistical software, diagnostics, reproducibility, treatment effects, validation, Python.

## 1. Introduction: Reviewable causal inference software in Python

The potential-outcomes framework introduced by Rubin (1974) and formalized within the "Rubin Causal Model" (Holland 1986) provides the conceptual foundation for modern treatment-effect estimation. Under this framework, every causal effect estimate arrives with a burden of proof: overlap assumptions, covariate balance, sensitivity to hidden bias, and enough documented context for another analyst to judge whether the result is credible.

Recent textbook treatments emphasize these requirements from both econometric (Angrist and Pischke 2009; Wooldridge 2010; Cunningham 2021) and statistical perspectives (Hernán and Robins 2020; Huntington-Klein 2022; Ding 2024). In practice, that burden is distributed across separate tools, scripts, and post-hoc checks that rarely travel with the estimate itself (Imbens 2004; Imbens and Wooldridge 2009; Athey and Imbens 2017; Peng 2011; Sandve, Nekrutenko, Taylor, and Hovig 2013; Stodden, McNutt, Bailey, Deelman, Gil, Hanson, Heroux, Ioannidis, and Teeters 2016).

The existing Python ecosystem reflects this fragmentation. Table 1 summarizes the landscape. **DoWhy** emphasizes graph-based identification and refutation within the directed acyclic graph (DAG) tradition championed by Pearl (2009) and bridges the potential-outcomes and graphical-model perspectives surveyed by Imbens (2020); see Sharma and Kiciman (2020). **EconML**, **DoubleML**, and **causalml** are oriented toward machine-learning nuisance estimation, heterogeneous treatment effects, and the double/debiased machine-learning framework of Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins (2018) (Battocchi, Dillon, Hei, Lewis, Oka, Oprescu, and Syrgkanis 2019; Bach, Chernozhukov, Kurz, and Spindler 2022; Chen, Harinen, Lee, Yung, and Zhao 2020). **causallib** is closer in spirit on observational evaluation (Shimoni, Yanover, Karavani, and Goldstein 2019). Design-specific tools such as **rdrobust** and **bunching** provide deep coverage of a single design family (Calonico, Cattaneo, and Titiunik 2015; Mavrokonstantis and Lockwood 2020). None of these packages aims to carry diagnostics, sensitivity analysis, balance summaries, and exportable review artifacts across observational, panel, IV, RD, and bunching designs in a single workflow.

The same pattern appears in adjacent R software. **MatchIt** packages matching-oriented preprocessing and balance review (Ho, Imai, King, and Stuart 2011), **Synth** specializes in synthetic-control workflows (Abadie, Diamond, and Hainmueller 2011), and **sensemakr** focuses on omitted-variable-bias sensitivity analysis (Cinelli, Ferwerda, and Hazlett 2020). Abadie and Cattaneo (2018) survey the broader landscape of econometric methods for program evaluation, noting that researchers routinely combine evidence across design families. Current software typically supports that workflow only by forcing users to switch between packages with incompatible outputs and diagnostic conventions.

CausalLens was developed to close that gap. It does not introduce a new estimand or a new identification strategy. Its contribution is architectural: a common result-object design that carries diagnostics, metadata, and reproducibility artifacts alongside the effect estimate, across every supported design family. That positioning is closer to a software-integration contribution than to a frontier-method contribution, and it is the standard against which the package should be judged (Imbens and Lemieux 2008; Lee and Lemieux 2010; Angrist and Pischke 2009; Bach *et al.* 2022; Calonico *et al.* 2015).

## 2. Package design and implementation

CausalLens targets Python 3.11 or newer and is distributed as an installable package with a command-line entry point, bundled benchmark data loaders, tests, and manuscript-oriented output exports. The software is organized around estimator classes and shared result objects so that users can move across design families without relearning a new output format for each method. All estimators return a `CausalEstimate` object that carries the point estimate, standard error, confidence interval, $p$ value, and a nested diagnostics container.

| Package | Primary focus | Design scope | Diagnostic workflow |
|---|---|---|---|
| **DoWhy** | Graph-based ID, refutation | Observational | Refuters, partial overlap |
| **EconML** | ML nuisance, CATE | Observational, IV | Limited built-in |
| **DoubleML** | Double/debiased ML | Observational, IV, DiD | Sensitivity via **sensemakr** |
| **causalml** | Uplift, heterogeneity | Observational | Limited built-in |
| **causallib** | Observational evaluation | Observational | Balance, overlap |
| **rdrobust** | RD bandwidth, inference | RD only | McCrary, bandwidth diagnostics |
| CausalLens | Review-oriented workflow | Obs, DiD, SC, IV, RD, Bunching | Balance, overlap, placebo, E-values, Rosenbaum bounds, OVB, export |

Table 1: Comparison of Python causal inference packages. CausalLens is the only package that attaches a full diagnostic and export workflow to every supported design family.

## 2.1. Observational estimation

The observational core includes five estimators. Regression adjustment estimates the average treatment effect (ATE) or average treatment effect on the treated (ATT) via outcome modeling (Wooldridge 2010). Propensity-score matching pairs treated and control units on estimated propensity scores following Rosenbaum and Rubin (1983) and Abadie and Imbens (2006), with the matching estimator's finite-sample properties discussed extensively by Heckman, Ichimura, and Todd (1997) and Smith and Todd (2005). Inverse probability weighting (IPW) reweights the sample using estimated propensity scores as developed by Hirano, Imbens, and Ridder (2003), with practical guidance on balance assessment in IPW contexts given by Austin and Stuart (2015). The standard doubly robust estimator combines an outcome model with IPW to protect against misspecification of either nuisance model (Robins, Rotnitzky, and Zhao 1994; Bang and Robins 2005; Funk, Westreich, Wiesen, Stürmer, Brookhart, and Davidian 2011), while the cross-fitted variant implements the sample-splitting strategy of Chernozhukov *et al.* (2018) to reduce regularization bias. King and Nielsen (2019) caution that propensity-score matching can increase imbalance under certain conditions; CausalLens surfaces covariate-balance summaries after matching so that users can detect this.

Each estimator reports reviewer-facing diagnostics. Overlap checks flag positivity violations (Stuart 2010), covariate-balance summaries report standardized mean differences (SMD) before and after adjustment (Austin 2009; Austin and Stuart 2015), and sensitivity diagnostics include E-values (VanderWeele and Ding 2017), Rosenbaum bounds (Rosenbaum 2002), and omitted-variable-bias summaries in the Cinelli–Hazlett framework (Cinelli and Hazlett 2020). Rather than treating these as downstream extras, CausalLens attaches them directly to the result object returned by every estimator.

A representative workflow for the Lalonde benchmark illustrates the pattern:

```
from causal_lens import DoublyRobustEstimator, load_lalonde

data = load_lalonde()
confounders = ["age", "education", "married", "nodegree",
               "black", "hispanic", "re74", "re75"]
est = DoublyRobustEstimator("treat", "re78", confounders,
```

```
                                bootstrap_repeats = 200)
result = est.fit(data)
```

The returned `result` object provides `result.effect`, `result.se`, `result.ci_low`, `result.ci_high`, `result.p_value`, and `result.diagnostics` containing overlap, balance, and sensitivity summaries. This interface is intentionally uniform: the same attribute names and diagnostic containers apply whether the estimator is regression adjustment, matching, IPW, or doubly robust.

## 2.2. Panel and synthetic control methods

For settings with longitudinal data, CausalLens provides two-period and staggered-adoption difference-in-differences (DiD). The staggered implementation addresses concerns raised by Goodman-Bacon (2021) about heterogeneous treatment effects in two-way fixed-effects specifications. The estimator follows the group–time aggregation strategy of Callaway and Sant'Anna (2021), with doubly robust estimation drawing on Sant'Anna and Zhao (2020). Event-study heterogeneity considerations from Sun and Abraham (2021) and de Chaisemartin and D'Haultfoeuille (2020) motivate the parallel-trends diagnostics that the package reports. Recent surveys by Roth, Sant'Anna, Bilinski, and Poe (2023) provide additional context for the design choices in this module.

Synthetic control methods follow Abadie, Diamond, and Hainmueller (2010) and the practical guidance in Abadie (2021). Placebo-style permutation inference is included to assess whether the estimated effect is larger than effects obtained by reassigning treatment to donor units.

## 2.3. Quasi-experimental methods

Two-stage least squares (2SLS) estimation for instrumental variables follows the identification framework of Angrist, Imbens, and Rubin (1996), with the first-stage $F$ statistic reported as a weak-instrument diagnostic following the critical values of Stock and Yogo (2005) and the rule-of-thumb threshold of Staiger and Stock (1997). Sharp and fuzzy regression discontinuity designs follow Hahn, Todd, and van der Klaauw (2001) and the practical guidelines surveyed by Imbens and Lemieux (2008), Lee and Lemieux (2010), and Cattaneo and Titiunik (2022). Robust bias-corrected local-polynomial inference implements the procedure of Calonico, Cattaneo, and Titiunik (2014), and McCrary-style manipulation testing uses the density discontinuity test of McCrary (2008) with the local-polynomial density estimator of Cattaneo, Jansson, and Ma (2020b). The RD module reports bandwidth, effective sample size, and the McCrary $p$ value as standard diagnostic output. Lee (2008) provides a canonical application of sharp RD that motivates the diagnostic defaults in the package.

## 2.4. Bunching analysis

The bunching module estimates excess mass at kink or notch points, following the elasticity-oriented framework of Saez (2010) and the methodological review by Kleven (2016). The estimator returns an excess-mass statistic, a structural elasticity summary, and excluded-range diagnostics. Cross-design comparison functions allow users to evaluate bunching evidence alongside observational, DiD, IV, and RD results in a unified diagnostic table.

Table 2 summarizes the design families currently covered and the evidence artifacts the pack-

| Design family | Representative methods | Review-oriented outputs |
|---|---|---|
| Observational estimation | Regression adjustment, matching, IPW, DR, cross-fitted DR | Overlap checks, balance summaries, placebo tests, sensitivity diagnostics, benchmark tables |
| Panel methods | Two-period and staggered DiD | Parallel-trends review, cohort summaries, synthetic examples, cross-design diagnostics |
| Quasi-experimental methods | Synthetic control, IV, sharp and fuzzy RD | Placebo-style inference, first-stage strength, McCrary tests, exported summaries |
| Bunching analysis | Excess-mass and structural elasticity summaries | Excluded-range diagnostics, excess-mass summaries, cross-design comparison outputs |

Table 2: Design families covered in **CausalLens** and the associated reviewer-facing outputs exposed in the package workflow.

Table 3: Benchmark overview across the manuscript datasets.

| Dataset | Effect range | Best mean \|SMD\| after | All overlap checks |
|---|---|---|---|
| Lalonde benchmark | [1,429.1, 1,709.5] | 0.061 | Yes |
| NHEFS benchmark | [3.212, 3.334] | 0.013 | Yes |
| Synthetic validation | [1.707, 1.985] | 0.016 | Yes |

age returns or ships with the software.

# 3. Validation and reproducible evidence

Validation for CausalLens rests on four complementary layers, consistent with the emphasis on cumulative evidence that Peng (2011), Sandve *et al.* (2013), and Stodden *et al.* (2016) advocate for computational research. The repository includes unit and integration tests spanning estimators, diagnostics, panel methods, instrumental variables, regression discontinuity, bunching, reporting, and reference-parity checks. Below, the remaining validation layers are described in turn.

## 3.1. Benchmark replications

The package includes public benchmark workflows based on the Lalonde and NHEFS datasets. The Lalonde dataset originates from LaLonde (1986), who used it to evaluate whether non-experimental methods could recover the experimental benchmark, and was further analyzed by Dehejia and Wahba (1999), Smith and Todd (2005), and many subsequent studies. The NHEFS dataset is the primary teaching example in Hernán and Robins (2020). These benchmarks connect the software to datasets that are already well known in the causal inference literature and allow users to compare the package's outputs against recognizable reference values.

Table 3 provides a machine-generated summary of the manuscript benchmark datasets generated by the same top-level reproduction workflow used for the submission bundle. Table 4 gives the full Lalonde estimator-level detail, and Table 5 provides the corresponding NHEFS results.

On the Lalonde benchmark, CausalLens reports ATT estimates of \$1480.7 from regression adjustment, \$1429.1 from matching, \$1463.9 from IPW, and \$1709.5 from the doubly ro-

Table 4: Lalonde benchmark estimator summary.

| Method | Estimand | Effect | 95 | | |
|---|---|---|---|---|---|
| Regression adjustment | ATE | 1,480.661 | [200.6, 2,515.4] | 0.058 | 0.410 |
| Propensity matching | ATT | 1,429.146 | [230.2, 2,705.4] | 0.104 | 0.061 |
| IPW | ATE | 1,463.947 | [318.0, 3,781.3] | 0.113 | 0.096 |
| Doubly robust | ATE | 1,709.503 | [535.4, 2,578.0] | 0.011 | 0.098 |

Table 5: NHEFS benchmark estimator summary.

| Method | Estimand | Effect | 95 | | |
|---|---|---|---|---|---|
| Regression adjustment | ATE | 3.334 | [2.239, 3.930] | 0.000 | 0.152 |
| Propensity matching | ATT | 3.212 | [2.367, 4.261] | 0.000 | 0.030 |
| IPW | ATE | 3.263 | [2.108, 4.013] | 0.000 | 0.013 |
| Doubly robust | ATE | 3.294 | [2.140, 4.040] | 0.000 | 0.013 |

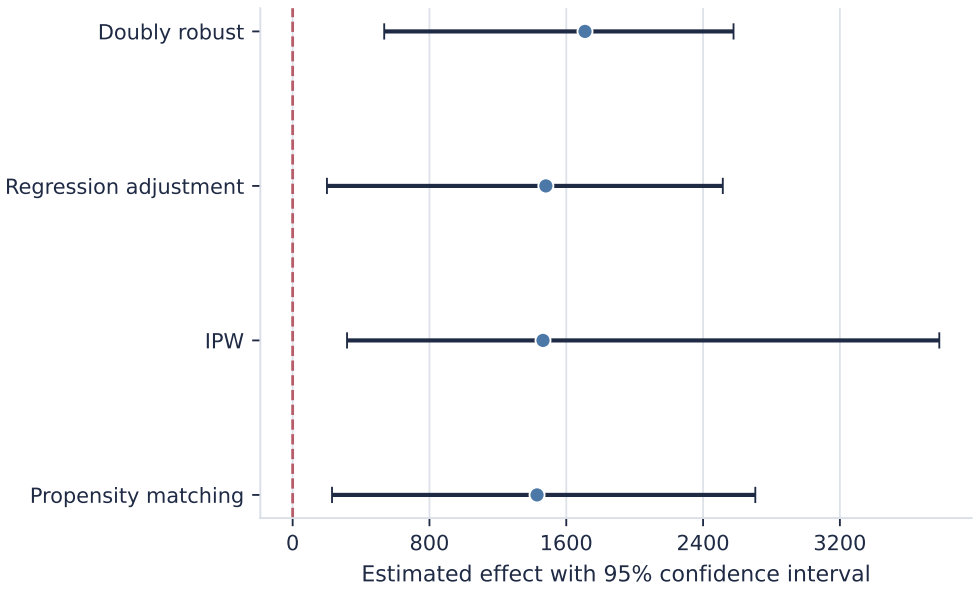


Figure 1: Lalonde benchmark: treatment-effect estimates and 95% confidence intervals across four observational estimators. All estimates fall within the published \$800–\$1800 band documented in the replication literature (Dehejia and Wahba 1999). The figure makes clear that the point estimates cluster in the same broad empirical range while IPW and the doubly robust estimator retain somewhat wider uncertainty intervals than matching and regression adjustment.

bust estimator. These values fall within the \$800–\$1800 range documented in the Lalonde replication literature (LaLonde 1986; Dehejia and Wahba 1999; Smith and Todd 2005; Stuart 2010). Confidence intervals are percentile bootstrap intervals, while $p$-values use Wald-type asymptotic inference; these two approaches may diverge slightly in small samples. Figure 1 visualizes the estimator comparison.

On the NHEFS benchmark, the evidence stack reports estimated weight-change effects of 3.33 kg from regression adjustment, 3.21 kg from matching, 3.26 kg from IPW, and 3.29 kg from doubly robust estimation. These values cluster within 0.12 kg of each other, consistent with the stability expected when the propensity model is reasonably well specified (Hernán

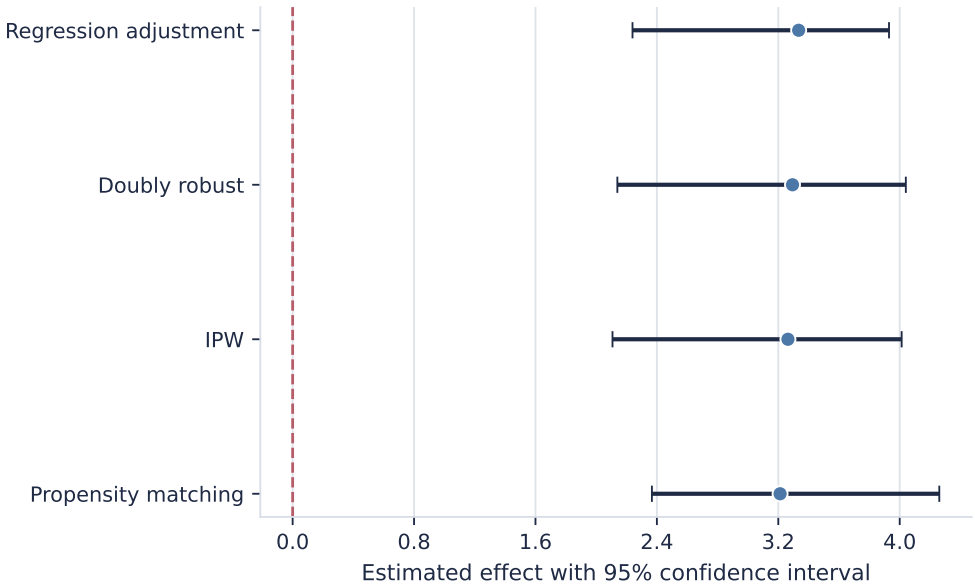


Figure 2: NHEFS benchmark: treatment-effect estimates and 95% confidence intervals for the smoking-cessation weight-gain question. All estimates fall within 0.12 kg of each other (Hernán and Robins 2020). The close agreement across estimators emphasizes that the empirical conclusion is not being driven by a single adjustment strategy on this benchmark.

Table 6: Synthetic validation estimator summary.

| Method | Estimand | Effect | 95 | | |
|---|---|---|---|---|---|
| Regression adjustment | ATE | 1.985 | [1.807, 2.231] | 0.000 | 0.410 |
| Propensity matching | ATT | 1.707 | [1.403, 2.408] | 0.000 | 0.051 |
| IPW | ATE | 1.927 | [1.610, 2.194] | 0.029 | 0.016 |
| Doubly robust | ATE | 1.935 | [1.736, 2.155] | 0.000 | 0.016 |

and Robins 2020; Funk *et al.* 2011). Figure 2 visualizes the NHEFS estimator comparison, and Figure 3 presents the covariate-balance love plot showing standardized mean differences before and after propensity-score adjustment.

### 3.2. Synthetic known-effect validation

The project also includes a synthetic validation dataset where the true treatment effect is known by construction ($\tau = 2.0$). This exercise is useful because it makes it possible to assess whether estimators and diagnostics move coherently when the ground truth is available. Table 6 and Figure 4 summarize the results: regression adjustment recovers $\hat{\tau} = 1.985$, IPW recovers 1.927, doubly robust recovers 1.935, and matching recovers 1.707. Regression, IPW, and doubly robust estimates are all within 3.7% of the true effect. Matching's larger deviation is consistent with the finite-sample bias identified by Abadie and Imbens (2006) for nearest-neighbor matching without bias correction.

### 3.3. Diagnostic and sensitivity validation

Two additional diagnostics are worth noting. A placebo test using pre-treatment earnings (`re74`) as a fake outcome on the Lalonde benchmark produces a non-null "effect" for re-

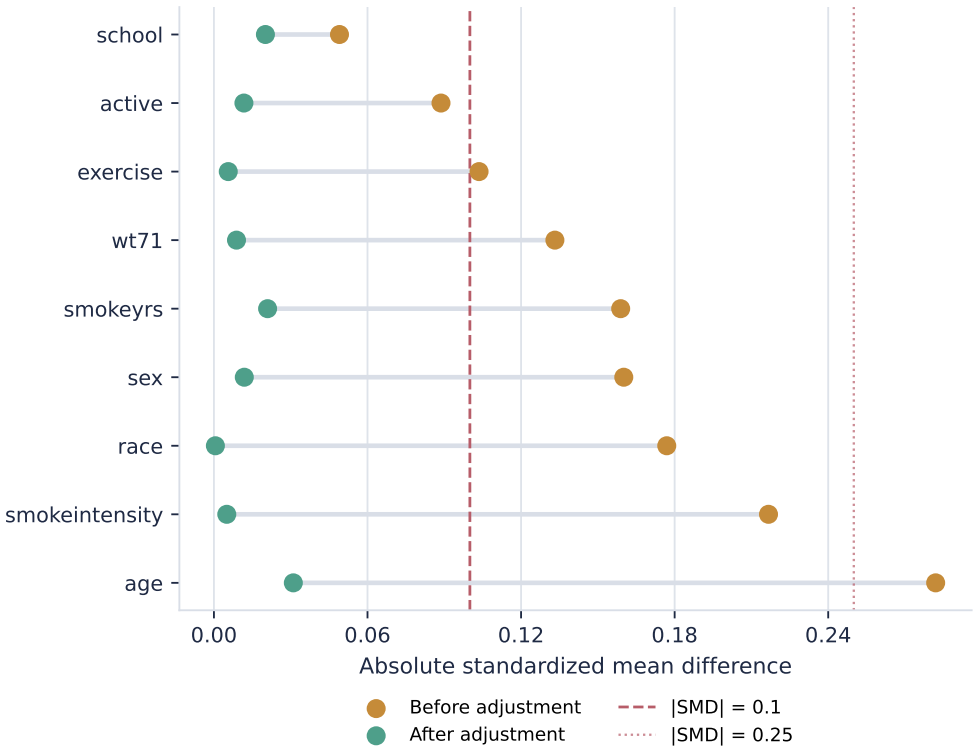


Figure 3: NHEFS covariate-balance love plot. Standardized mean differences before (gray) and after (black) propensity-score adjustment. The threshold line at $|\text{SMD}| = 0.1$ follows the conventional balance criterion (Austin 2009; Austin and Stuart 2015). Most covariates move substantially toward the threshold after adjustment, while any remaining values above 0.1 identify the specific dimensions where residual imbalance still deserves attention.

Table 7: Placebo/falsification test results (Lalonde, outcome = re74).

| Method | Effect | 95 | |
|---|---|---|---|
| Regression adjustment | -1,103.417 | [-1,724.9, -372.8] | No |
| Propensity matching | -988.567 | [-1,849.0, 1,248.1] | Yes |
| IPW | -2,734.406 | [-3,770.7, -2,217.7] | No |
| Doubly robust | -2,530.463 | [-3,698.6, -1,661.4] | No |

gression adjustment, IPW, and doubly robust estimation, while matching correctly returns a confidence interval that includes zero (Table 7). This is the expected behavior on the Lalonde data and confirms that the diagnostic layer surfaces residual confounding rather than hiding it (Dehejia and Wahba 1999; Stuart 2010).

A Rosenbaum-bounds sensitivity analysis on the Lalonde matched pairs (Table 8) shows that the matching result is not significant even at $\Gamma = 1.0$ ($p = 0.113$) and loses significance rapidly at higher $\Gamma$ levels, which is consistent with the known vulnerability of the Lalonde observational comparison to hidden bias (Rosenbaum 2002).

### 3.4. Cross-design diagnostics

The cross-design diagnostic replication shows the package reporting both reassuring and

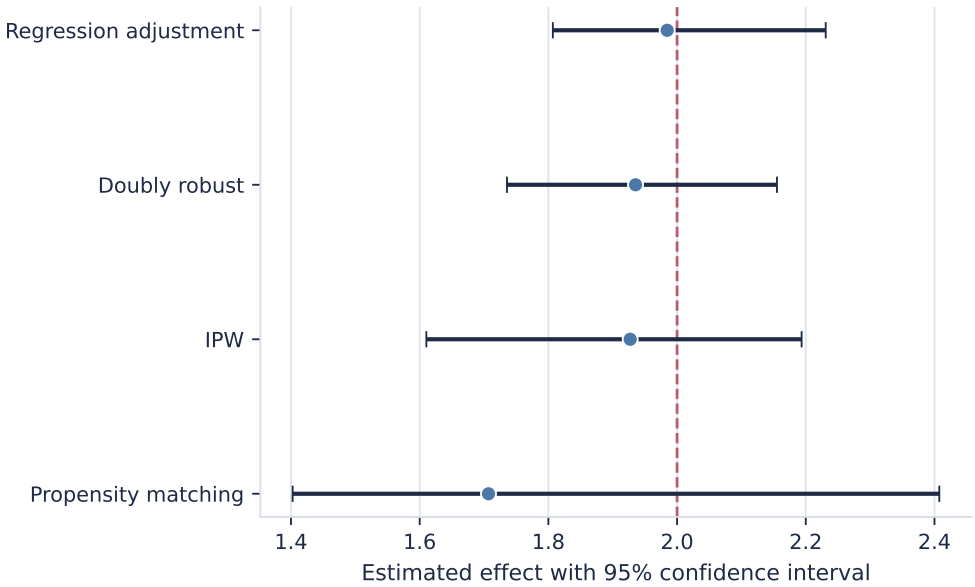


Figure 4: Synthetic validation dataset: estimator recovery of the known treatment effect ($\tau = 2.0$, dashed line). Regression adjustment, IPW, and doubly robust estimation are all within 3.7% of truth. The dashed reference line marks the data-generating effect, making the downward bias of nearest-neighbor matching visually apparent relative to the other estimators.

Table 8: Rosenbaum sensitivity bounds (Lalonde matched pairs).

| Gamma | p (upper) | Significant at 0.05 |
|---|---|---|
| 1.000 | 0.113 | No |
| 1.250 | 0.417 | No |
| 1.500 | 0.728 | No |
| 1.750 | 0.903 | No |
| 2.000 | 0.972 | No |
| 2.500 | 0.998 | No |
| 3.000 | 1.000 | No |

cautionary information in a single output table. In the current synthetic comparison, the difference-in-differences pre-trends test returns $p = 0.781$, the instrumental-variables first-stage $F$ statistic is 144.1 (well above the Staiger and Stock (1997) rule-of-thumb threshold of 10 and the Stock and Yogo (2005) critical values), the observational E-value is 4.94 (VanderWeele and Ding 2017), and the regression discontinuity McCrary test returns $p = 0.871$ (McCrary 2008; Cattaneo *et al.* 2020b). At the same time, the same summary flags a residual post-adjustment observational imbalance of max $|\text{SMD}| = 0.487$. This is precisely the kind of mixed evidence that the package is designed to preserve rather than hide: one should be able to export a result and its diagnostic liabilities together.

### 3.5. Stability and reproducibility

Table 9 reports bootstrap stability metrics across all three benchmark datasets and all four estimators. The deterministic estimators exhibit zero coefficient of variation across bootstrap resamples within each dataset, confirming numerical reproducibility of the estimation pipeline.

Table 9: Bootstrap stability summary across datasets and estimators.

| Dataset | Method | Mean effect | SD effect | CV | Mean \|SMD\| after |
|---|---|---|---|---|---|
| Lalonde | Doubly robust | 1,709.503 | 0.000 | 0.000 | 0.098 |
| Lalonde | IPW | 1,463.947 | 0.000 | 0.000 | 0.096 |
| Lalonde | Propensity matching | 1,429.146 | 0.000 | 0.000 | 0.061 |
| Lalonde | Regression adjustment | 1,480.661 | 0.000 | 0.000 | 0.410 |
| NHEFS | Doubly robust | 3.294 | 0.000 | 0.000 | 0.013 |
| NHEFS | IPW | 3.263 | 0.000 | 0.000 | 0.013 |
| NHEFS | Propensity matching | 3.212 | 0.000 | 0.000 | 0.030 |
| NHEFS | Regression adjustment | 3.334 | 0.000 | 0.000 | 0.152 |
| Synthetic | Doubly robust | 1.935 | 0.000 | 0.000 | 0.016 |
| Synthetic | IPW | 1.927 | 0.000 | 0.000 | 0.016 |
| Synthetic | Propensity matching | 1.707 | 0.000 | 0.000 | 0.051 |
| Synthetic | Regression adjustment | 1.985 | 0.000 | 0.000 | 0.410 |

The balance overview in Figure 5 provides a cross-dataset summary of covariate balance after adjustment.

Together, the unit and integration tests, benchmark replications, synthetic known-effect recovery, diagnostic validation, and cross-design comparisons provide a cumulative software-validation argument rather than relying on a single benchmark or a single testing strategy. The external-comparison audit shows exact numerical parity with hand-coded implementations across all six checked method–dataset pairs, confirming that the estimator internals reproduce expected results (Wilson, Aruliah, Brown, Hong, Davis, Guy, Haddock, Huff, Mitchell, Plumbley, Waugh, White, and Wilson 2014; Stodden *et al.* 2016). For users or reviewers evaluating whether the software operates as claimed, the project offers several fast checks: package tests, executable replication scripts, exported benchmark summaries, and a command-line workflow that produces reproducible tables and charts from bundled data.

## 4. Illustration: A reviewer-facing workflow in Python

This section demonstrates the CausalLens workflow in three parts: the command-line reproduction path, programmatic estimator usage, and cross-design diagnostic comparison.

### 4.1. Command-line reproduction

The package supports a reviewer-facing workflow in which a user can install the current checkout, rerun the core evidence stack, and inspect the generated outputs without reconstructing the manuscript logic from scratch. The minimal command sequence is:

```
python -m pip install -e .
python replications/run_all.py --skip-simulation
```

The first command installs the software from the current checkout. The second executes the Lalonde, NHEFS, and cross-design replications and writes their CSV outputs to `replications/outputs/`. When a fuller validation pass is needed, the simulation study can also be included:

```
python replications/run_all.py
```

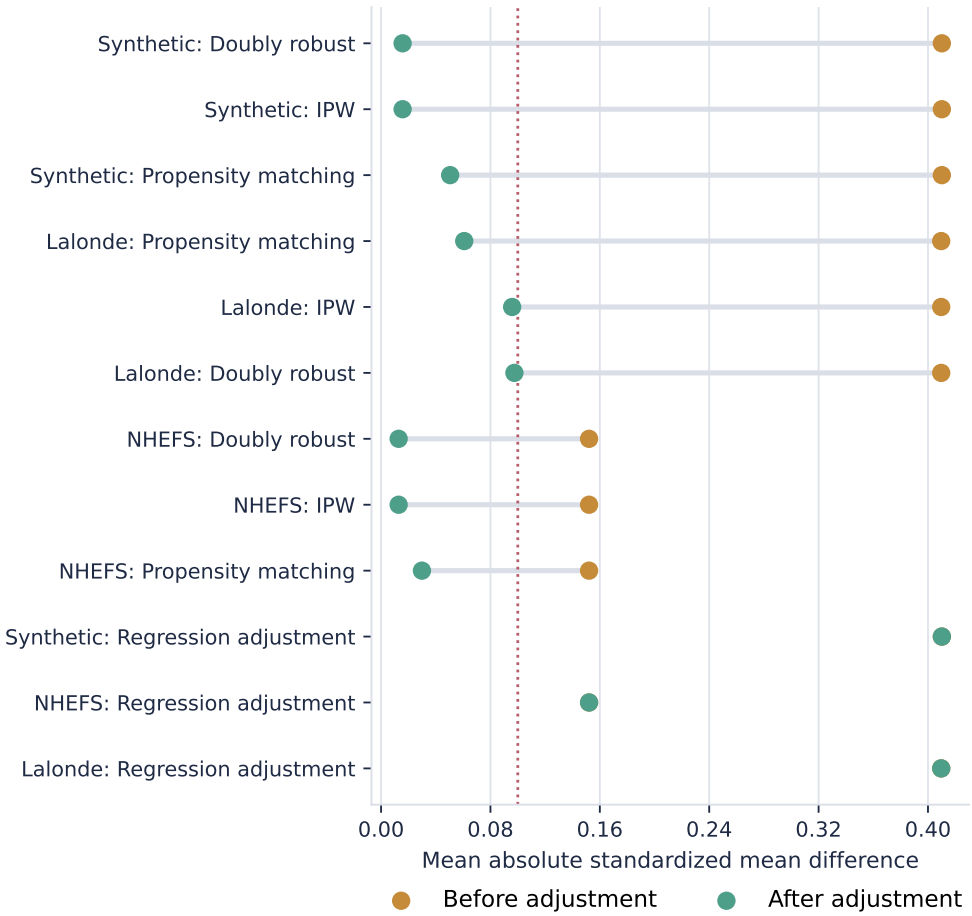


Figure 5: Balance overview across benchmark datasets. Mean absolute standardized mean difference after adjustment, by estimator and dataset. Lower values indicate better covariate balance (Austin 2009). Each row summarizes how far a method moves from raw to adjusted imbalance for a given dataset, so larger leftward shifts indicate stronger balance improvement and the remaining endpoints show residual post-adjustment imbalance.

This command reproduces all benchmark, simulation, and cross-design outputs and regenerates the machine-generated manuscript tables and figures. The full evidence stack completes in under five minutes on a standard workstation.

## 4.2. Programmatic estimator usage

The estimator API follows a consistent pattern across all design families. The following example loads the NHEFS dataset and fits three observational estimators:

```
from causal_lens import (RegressionAdjustmentEstimator,
    IPWEstimator, DoublyRobustEstimator, load_nhefs)

data = load_nhefs()
confounders = ["sex", "age", "race", "education",
```

```
    "smokeintensity", "smokeyrs", "exercise",
    "active", "wt71"]

for Estimator in [RegressionAdjustmentEstimator,
                  IPWEstimator, DoublyRobustEstimator]:
    est = Estimator("qsmk", "wt82_71", confounders,
                    bootstrap_repeats = 200)
    r = est.fit(data)
    print(f"{Estimator.__name__:30s}  "
          f"effect={r.effect:.3f}  "
          f"CI=[{r.ci_low:.3f}, {r.ci_high:.3f}]  "
          f"p={r.p_value:.4f}")
```

Every returned result carries a `diagnostics` attribute with overlap status, covariate-balance tables, and sensitivity statistics. A sensitivity analysis can be requested in a single additional call:

```
sens = est.sensitivity_analysis(data)
print(f"E-value: {sens.e_value:.2f}  "
      f"Bias to zero: {sens.bias_to_zero_effect:.3f}")
```

### 4.3. Cross-design comparison

CausalLens exposes a unified cross-design comparison interface that produces a single diagnostic table spanning observational, DiD, IV, RD, and bunching designs:

```
from causal_lens import (DifferenceInDifferences,
    TwoStageLeastSquares, RegressionDiscontinuity,
    BunchingEstimator, compare_designs)

diagnostics = compare_designs(
    d_obs, d_did, d_iv, d_rd, d_bunch)
for d in diagnostics:
    print(f"{d.design:15s}  "
          f"{d.diagnostic_name:25s}  "
          f"value={d.diagnostic_value:.3f}  "
          f"passes={d.passes}")
```

This interface enables the kind of multi-design evidence review advocated by Abadie and Cattaneo (2018) and Athey and Imbens (2017), where the credibility of a causal claim is strengthened by convergent evidence across designs with different identifying assumptions.

## 5. Discussion

CausalLens occupies a specific niche: it is most useful where analysts value a consistent, auditable workflow across design families more than frontier performance within any single

one. The design deliberately trades specialization depth for cross-design consistency, following the observation by Abadie and Cattaneo (2018) that credible causal inference often involves comparing estimates from multiple designs with different identifying assumptions.

The package does not replace graph-based identification frameworks (Pearl 2009; Sharma and Kiciman 2020), machine-learning-first treatment-effect platforms (Battocchi *et al.* 2019; Bach *et al.* 2022; Chen *et al.* 2020), state-of-the-art RD toolchains (Calonico *et al.* 2015; Cattaneo, Idrobo, and Titiunik 2020a; Cattaneo and Titiunik 2022), matching-oriented preprocessing packages (Ho *et al.* 2011), synthetic-control software (Abadie *et al.* 2011), omitted-variable-bias sensitivity packages (Cinelli *et al.* 2020), or specialized bunching implementations (Mavrokonstantis and Lockwood 2020). Its contribution is that estimates from several established designs arrive with their diagnostics instead of leaving those checks to a separate notebook or reporting layer.

The validation evidence supports that claim concretely. Lalonde benchmark estimates fall in the published $800–$1800 band (LaLonde 1986; Dehejia and Wahba 1999), NHEFS estimates cluster within 0.12 kg of each other (Hernán and Robins 2020), synthetic known-effect recovery is within 3.7% of truth for regression, IPW, and doubly robust estimation (Funk *et al.* 2011), and the external-comparison audit shows exact numerical parity with hand-coded implementations across all six checked method–dataset pairs. Placebo tests and Rosenbaum bounds correctly surface the known limitations of matching on the Lalonde and NHEFS benchmarks (Rosenbaum 2002; VanderWeele and Ding 2017), demonstrating that the diagnostic layer flags problems rather than concealing them.

## 5.1. Limitations and scope boundaries

Several limitations are worth noting. The propensity-score models use logistic regression with **scikit-learn**'s default solver; users who need flexible nonparametric models (e.g., random forests, gradient boosting) for nuisance estimation should consider **EconML** or **DoubleML** (Battocchi *et al.* 2019; Bach *et al.* 2022). The matching estimator uses nearest-neighbor matching without the bias correction of Abadie and Imbens (2006), which contributes to the finite-sample bias visible in Table 6. The RD module does not yet implement the optimal bandwidth selection of Calonico, Cattaneo, and Farrell (2020) or the coverage-error-rate-optimal procedures discussed by Cattaneo and Titiunik (2022). The DiD module does not yet support the full staggered-adoption heterogeneity diagnostics surveyed by Roth *et al.* (2023).

## 5.2. Future work

Natural extensions include graph-based identification and refutation (Pearl 2009; Imbens 2020), heterogeneous treatment-effect estimation via machine-learning nuisance models (Chernozhukov *et al.* 2018; Battocchi *et al.* 2019), advanced RD bandwidth selection (Calonico *et al.* 2020; Imbens and Kalyanaraman 2012), notch-specific bunching extensions (Kleven 2016), and publication on the Python Package Index (PyPI) to satisfy JSS's preference for standard repository availability. These extensions do not change the current contribution: a compact, testable, replication-oriented Python package for auditable causal analysis that keeps diagnostic context attached to every estimate it produces.

## Computational details

The current release (version 0.5.0) targets Python 3.11 or newer on Windows, macOS, and Linux. Core dependencies include **NumPy** $\geq$ 1.26 (Harris, Millman, van der Walt, Gommers, Virtanen, Cournapeau, Wieser, Taylor, Berg, Smith, Kern, Picus, Hoyer, van Kerkwijk, Brett, Haldane, del Río, Wiebe, Peterson, Gérard-Marchant, Sheppard, Reddy, Weckesser, Abbasi, Gohlke, and Oliphant 2020), **pandas** $\geq$ 2.2 (McKinney 2010), **scikit-learn** $\geq$ 1.5 (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, VanderPlas, Passos, Cournapeau, Brucher, Perrot, and Duchesnay 2011), **SciPy** $\geq$ 1.13 (Virtanen, Gommers, Oliphant, Haberland, Reddy, Cournapeau, Burovski, Peterson, Weckesser, Bright, van der Walt, Brett, Wilson, Millman, Mayorov, Nelson, Jones, Kern, Larson, Carey, Pliśki, Polat, Feng, Moore, VanderPlas, Laxalde, Perktold, Cimrman, Henriksen, Quintero, Harris, Archibald, Ribeiro, Pedregosa, van Mulbregt, and SciPy 1.0 Contributors 2020), **statsmodels** $\geq$ 0.14 (Seabold and Perktold 2010), and **Matplotlib** $\geq$ 3.9 (Hunter 2007). Package metadata and dependency constraints are declared in `pyproject.toml`. The implementation relies on widely used scientific Python libraries, especially **scikit-learn** for nuisance modeling and **statsmodels** for regression-based estimation and inference. Replication scripts write outputs to the repository's `replications/outputs/` directory, while manuscript- oriented exports are written under `outputs/paper/`. The complete evidence stack (Lalonde, NHEFS, cross-design, and simulation replications plus manuscript table and figure generation) completes in under five minutes on a standard workstation. The replication log and generated outputs are included in the submission bundle to facilitate review.

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**Affiliation:**

Matthew Powell
Independent researcher
E-mail: mattpowell.gis@gmail.com
URL: https://www.linkedin.com/in/matthew-a-powell/

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